

PIXEL NOIR

Game Design Document (GDD) — Version 3.0

Genre: Top-Down Detective Adventure

Target Target: PC / Desktop

Status: Approved MVP Specification

1. VISION STATEMENT & DESIGN PILLARS

Pixel Noir is a cozy, top-down pixel art detective adventure where every mystery is logically constructed, every clue tells the truth, and every investigation presents a unique case to solve. The long-term architectural goal is a procedural mystery engine capable of generating infinitely replayable cases while maintaining strict logical solvability. The player does not grow stronger through physical combat, but through accumulated facts, refined investigative capabilities, and deductive precision.

1. Investigation Over Combat

No battles, weapons, or physical violence. Conflict is expressed through observation, dialogue, non-violent psychological encounters, and logical elimination.

2. Every Clue Is True

The underlying world simulation never lies. Physical clues exist for a deterministic reason. Misdirection arises only from personal NPC bias, incomplete claims, or player misinterpretation.

3. Knowledge & Skill Evolution As Progression

Progression stems from expanding facts on the dynamic Logic Grid and organically honing investigative skills through direct activity rather than point allocation.

4. Every NPC Has Something To Hide

Every town resident maintains personal secrets, motives, and daily schedules. Many secrets act as authentic red herrings unrelated to the main crime.

5. High-Stakes Deduction

Deduction is active gameplay. Players manually manage matrix hypotheses, cross-reference data, and accept the risk of misjudging evidence or executing incorrect arrests.

6. Replayability Through Logic

Procedural case design built around verifiable logical paths, ensuring every case seed is 100% solvable without guessing.

2. CORE GAMEPLAY LOOP & REPUTATION SYSTEM



Fail State & Detective Reputation

Failing to solve a case within the deadline or botching high-stakes interrogations does **not** trigger a hard restart. Instead, it degrades your overall **Detective Reputation**. Reputation dynamically affects branched narrative paths, NPC willingness to cooperate, dialogue tones, and baseline composure during future interrogations.

3. TIME BUDGET, SCHEDULE & WORLD SYSTEMS

Granular Time Economy

Cases operate under a strict time deadline (e.g., 72 total hours). Time advances exclusively when executing deliberate player choices. Entering a room or walking around within a space incurs zero time cost; time is consumed strictly by discrete actions taken inside that space.

ACTION CATEGORY	SPECIFIC ACTION	TIME COST	NOTES
District Travel	Moving between town zones / buildings	15 Minutes	Flat cost for changing major locations
Room Entry	Stepping inside a building or room	0 Minutes	Free movement inside non-gated areas
Object Inspection	Examining evidence, physical items, room search	30 Minutes	Reduced to 15m via <i>Forensic Efficiency</i> perk
Casual Interview	Standard dialogue, general questions	30 Minutes	Gathers initial claims & unlocks schedule data
Interrogation	Structured Pokémon-style confrontation	60 Minutes	High-stakes encounter; requires preparation
Resting / Waiting	Deliberate waiting anywhere in the world	15 Min Increments	Universal mechanic to align with NPC schedules

Lighting Engine & NPC Schedules

- **Day/Night Transitions:** Ambient lighting smoothly shifts across Dawn, Day, Dusk, and Night.

- **World Gating:** Daytime enables access to public businesses and official offices; nighttime locks residential doors, opens illicit spots (alleyways, secret meetups), and alters crime scene visibility.
- **Dynamic Schedules:** NPCs follow strict hourly schedules, moving across locations throughout the day.

4. THE LOGIC GRID SYSTEM (DEDUCTION ENGINE)

Progressive Schema & Matrix Operations

- **Dynamic Categorization:** Matrix categories (Suspects, Locations, Crime Scenes, Motives, Timelines) start hidden and dynamically append rows/columns upon discovering new case angles.
- **Matrix Cell States:**
 - [✓] **Direct Affirmation:** Confirmed match between two attributes.
 - [x] **Direct Negation:** Confirmed elimination of an attribute pairing.
 - [?] **Unverified Claim:** Information derived from subjective NPC interviews.
- **Automatic Process of Elimination:** When all slots except one in a row/column are tagged [x], the remaining slot automatically resolves to [✓].

High-Stakes Manual Operations & Misclick Risk

Players are allowed to manually tag cells based on their own reasoning. The base UI does **not** automatically flag logical contradictions or prevent incorrect tags. Incorrect player assumptions can lead to building flawed cases. Automatic contradiction flagging is unlocked exclusively as a late-tier perk within the *Deception Sensitivity* skill tree.

Accusation Gate

To execute an arrest, the primary suspect's horizontal row on the Logic Grid must be completely resolved with confirmed links across all mandatory categories (Suspect + Crime Scene + Motive + Means).

5. NON-VIOLENT INTERROGATION SYSTEM (POKÉMON-STYLE)

Interrogations are structured, turn-based encounters inspired by classic battle systems. Player and suspect face off using psychological resources, tactical stances, and item support.

[SUSPECT]
Arthur Vance [??]
RESOLVE: [███████████]
STANCE: [Rattled]

[DETECTIVE]
Detective
COMPOSURE: [███████████]
ITEMS: [Double Espresso x1, Drip Coffee x2]

Text Log: Arthur shifts uncomfortably in his chair...

1. TACTICS

2. PRESENT EVIDENCE

3. CONSUME COFFEE

4. RETRY LATER (CONCEDE)

Resource Mechanics

- **Suspect Resolve (Enemy HP):** Suspect's psychological barrier. Reduced by effective tactics, probing questions, and presenting direct evidence that exposes lies. Reaching 0 Resolve results in a complete confession and verified grid updates.

- **Detective Composure (Player Stamina / HP):** Represents the detective's control over the interrogation. Misapplying evidence, falling for suspect traps, or taking aggressive counters depletes Composure. If Composure hits 0, the player loses control, incurs a **Reputation Hit**, and the suspect locks out further interviews until the next day.

4-Slot Action Menu Breakdown

1. Tactics (Standard Commands)

- **Press:** Direct attack dealing moderate Resolve damage. Backfires and reduces player Composure if suspect is currently *Guarded*.
- **Empathize:** Lowers suspect defense or reduces incoming Composure damage for 2 turns.
- **Observe:** Free scan action; reveals suspect's active Stance and highlights psychological weaknesses.

2. Present Evidence (Journal & Logic Grid Overlay)

Selecting *Present Evidence* slides the integrated **Clue Journal / Logic Grid** over the bottom half of the screen. Players select a specific entry to throw against the suspect's active claims:

- **WORLD FACT [!]** vs. **Lie:** Deals **Critical Hit** Resolve damage, breaks defensive stances, and permanently converts the claim cell [?] to a locked truth [✓].
- **Irrelevant Evidence:** Misses the target. Suspect counters, dealing Composure damage to the detective.

3. Consume Coffee (In-Battle Composure Recovery)

Restores Detective Composure during combat without forfeiting the entire turn structure.

COFFEE GRADE	COMPOSURE RESTORED	AVAILABILITY & COST
Instant Coffee	Low (+15% – 20%)	Abundant, cheap (Police Station, Diners)
Drip Coffee	Moderate (+40% – 50%)	Standard cost, available at town Cafes
Double Espresso	Maximum (+100%)	Rare, high-cost (Specialty Shops)

4. Retry Later / Concede

Allows the player to voluntarily end the encounter. Incurs a minor reputation penalty, but preserves remaining Composure and puts the suspect on a 24-hour cooldown before re-interrogating.

Suspect Psychological Stances

- **Guarded:** Takes 50% reduced Resolve damage from standard *Press* tactics. Requires presenting physical World Facts to break.
- **Rattled:** Takes 2x damage from *Press* tactics.
- **Cornered:** Suspect takes high Resolve damage, but their rebuttal counter-attacks deal increased Composure damage to the player.

6. ACTIVITY-BASED SKILL EVOLUTION

Skills level up **organically through activity** (e.g., conducting searches, questioning NPCs, cross-examining). Perks are automatically awarded upon reaching usage thresholds—no manual stat points or level-up menus.

SKILL DISCIPLINE	PRIMARY PROGRESSION ACTIVITY	AUTOMATIC PERKS UNLOCKED	GAMEPLAY IMPACT
Interrogation & Persuasion	Conducting interrogations & breaking suspect stances	Active Listening, Steel Nerve, Pressing Insight	Reduces composure loss, boosts resolve damage, highlights posture shifts.
Deception Sensitivity	Exposing lies & identifying claim conflicts	Micro-Expressions, Inconsistency Detection, Truth Sense	Flags unverified claims, highlights Logic Grid contradictions, reveals stance types.
Perception & Search	Inspecting objects & searching rooms	Eagle Eye, Keen Insight, Forensic Efficiency	Passively highlights hidden interactables; cuts object inspection time to 15m.
Navigation & Time	Traveling districts & operating during night hours	Street Smarts, Schedule Sense, Night Owl	Cuts district travel to 10m; reveals NPC schedule shifts 1 hour in advance.
Deductive Reasoning	Filling Logic Grid rows & cross-referencing clues	Epiphany, Cross-Examination, Auto-Elimination	Auto-highlights remaining grid options upon reaching logical thresholds.

7. MVP SCOPE SPECIFICATIONS

- **Setting:** Small compact town district featuring 8–12 fully scheduled NPCs and a dynamic day/night lighting loop.
- **Handcrafted Case:** 1 complete handcrafted initial mystery (Kidnapping / Missing Person) to validate time tracking, non-violent Pokémon interrogation, coffee recovery mechanics, dynamic grid expansion, and activity perk progression.
- **Core Deliverable Systems:**
 1. Interactive Logic Grid Matrix & Integrated Clue Journal Overlay.
 2. Discrete Time-Budget Tracker (72-hour clock, 15m/30m action costs, Rest action).
 3. Pokémon-Style Turn-Based Interrogation UI with Coffee inventory & Stance states.
 4. Reputation System tied to failure, timeout, and interrogation cooldown loops.
 5. Activity-based automatic skill perk triggers.